

AUTOMATIC MOMENT CONTROL FOR SELF-CONSISTENT OSCILLATOR BRANCHES

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ABSTRACT. For the finite-dimensional self-consistently renormalized oscillator of Baez and Zhou, local uniqueness was stated under a bounded interaction-moment hypothesis. We prove that this hypothesis follows from norm continuity of the ground-state branch. State-relative Wick polynomials form a formal multivariate Appell family. If a norm-convergent sequence has an escaping top moment, total-variation concentration makes all lower moments subhomogeneous, and changing the reference state forces the Wick expectation in the limit state to tend to $-\infty$. Ground-state minimality supplies an opposing uniform lower bound, including at zero coupling through the free vacuum. Thus the top moment is bounded on every compact coupling interval. The cone inverse theorem then yields a uniform neighborhood on which every shorter closed interval carries exactly one norm-continuous nonnegative Schwartz ground-state branch, without an a priori moment bound.

1. INTRODUCTION

Ordinary normal ordering originates in Wick's contraction formula [1]; Gaussian Wick polynomials and constructive field-theory implementations are treated, from complementary viewpoints, in [2–5]. The products in this paper are different: their reference state need not be Gaussian, and their centering and commutator axioms make them a state-relative formal Appell family.

This state-relative construction grew out of Segal's weak-process calculus and his construction of nonlinear local quantum processes [6–9]; a broader account is given by Baez, Segal, and Zhou [10]. Friedman isolated the corresponding renormalized oscillator equation and its translation obstruction [11]. Baez and Zhou then formulated finite-dimensional self-consistent Wick ordering as a nonlinear ground-state problem and proved a local branch theorem for coercive polynomial interactions [12].

A self-consistent ground-state equation, in which the operator depends on the state it is meant to produce, is the setting of Hartree-type problems, and the uniqueness of such ground states has its own literature, and in it the answers are conditional: Lenzmann settled the pseudo-relativistic Hartree equation for ground states of sufficiently small mass, and there for all but countably many values of it [13], while Griesemer and Hantsch settled the Hartree–Fock ground state of a closed-shell atom whose nuclear charge is large enough relative to its electron number [14]; the question remains active for nonlinear Hartree equations [15]. What distinguishes the present problem from all of these is that the nonlinearity enters through the Wick ordering rather than through a convolution, so the interaction moment that has to be controlled is the one the ordering itself produces.

The analytic-family and self-adjoint Schrödinger background is supplied by standard perturbation and operator theory [16–18]. Baez and Zhou's uniqueness argument assumes that the interaction moment remains bounded along the branch, and they record the status of that assumption themselves: they call it a technical condition and ask whether it can be omitted, while noting that continuity on the closed interval cannot be, since uniqueness already fails on a half-open one [12]. This paper answers that question. Since norm convergence alone does not normally control unbounded polynomial observables, the answer uses the self-consistent variational structure rather than a compactness slogan.

The mechanism is a sign contradiction. Formal Appell calculus identifies the leading behavior of a Wick polynomial when its reference state develops an escaping top moment. Total-variation convergence of position laws makes all lower-moment contributions negligible, leaving a negative coercive term. The same expectation, evaluated in a fixed comparison state, is bounded below by ground-state minimality. This also handles sequences of couplings approaching zero, where the free vacuum is the comparison state.

Theorem 1.1. *Let*

$$H_0 = \frac{1}{2} \sum_{j=1}^n \omega_j (p_j^2 + q_j^2 - 1), \quad \omega_j > 0,$$

on $L^2(\mathbb{R}^n)$, and let P be a real polynomial of even degree N satisfying $P(x) \geq \varepsilon|x|^N - k$ for some $\varepsilon, k > 0$. There is $\Lambda > 0$ such that, for every $0 < \lambda_0 \leq \Lambda$, exactly one normalized nonnegative Schwartz-valued norm-continuous map $u : [0, \lambda_0] \rightarrow L^2(\mathbb{R}^n)$ starts at the free vacuum and has $u(\lambda)$ as the ground state of the self-adjoint closure of

$$H_0 + \lambda : P(q) :_{u(\lambda)}.$$

In particular, no a priori bound on $\langle u(\lambda), P(q)u(\lambda) \rangle$ is required.

2. STATE-RELATIVE WICK POLYNOMIALS AS A FORMAL APPELL FAMILY

For a normalized Schwartz vector u write

$$\mu_\alpha(u) = \langle u, q^\alpha u \rangle, \quad A_\alpha^u(q) = : q^\alpha :_u. \quad (2.1)$$

The defining expectation and commutator axioms for state-relative renormalized products [12] are

$$A_0^u = 1, \quad \langle u, A_\alpha^u(q)u \rangle = 0 \quad (|\alpha| > 0), \quad \partial_{q_j} A_\alpha^u = \alpha_j A_{\alpha - e_j}^u. \quad (2.2)$$

Proposition 2.1. *Let*

$$\mathcal{A}_u(t, q) = \sum_{\alpha \geq 0} A_\alpha^u(q) \frac{t^\alpha}{\alpha!}, \quad M_u(t) = \sum_{\alpha \geq 0} \mu_\alpha(u) \frac{t^\alpha}{\alpha!} \quad (2.3)$$

as formal power series. Then

$$\mathcal{A}_u(t, q) = \frac{e^{t \cdot q}}{M_u(t)}. \quad (2.4)$$

Proof. The third axiom in (2.2) gives $\partial_{q_j} \mathcal{A}_u = t_j \mathcal{A}_u$, so $\mathcal{A}_u(t, q) = e^{t \cdot q} C_u(t)$ for a series C_u not involving q . Taking the u -expectation of (2.3) coefficientwise and using the second axiom gives $M_u(t) C_u(t) = 1$; since $M_u(0) = 1$, the formal reciprocal exists and is unique. $\square$

No analytic moment-generating function is asserted or needed: every identity here is an identity of formal series, and each coefficient is a finite algebraic expression in finitely many moments.

Proposition 2.2 (Change of reference state). *Let v be another normalized Schwartz vector and put $r_\alpha(v, u) = \langle v, A_\alpha^u(q)v \rangle$. Then*

$$\sum_{\alpha} r_\alpha(v, u) \frac{t^\alpha}{\alpha!} = \frac{M_v(t)}{M_u(t)}, \quad (2.5)$$

equivalently

$$r_\alpha(v, u) = \mu_\alpha(v) - \mu_\alpha(u) - \sum_{\substack{0 < \beta \leq \alpha \\ \beta \neq \alpha}} \binom{\alpha}{\beta} \mu_\beta(u) r_{\alpha-\beta}(v, u). \quad (2.6)$$

Proof. Take the v -expectation of (2.4) coefficientwise, which gives (2.5); equating coefficients in $M_u(t) \sum_{\alpha} r_\alpha t^\alpha / \alpha! = M_v(t)$ gives (2.6). $\square$

3. CONCENTRATION UNDER TOP-MOMENT ESCAPE

Lemma 3.1. *Let probability measures $\nu_j \rightarrow \nu$ in total variation and suppose $M_j = \int |x|^N d\nu_j \rightarrow \infty$. Then for every $0 < r < N$,*

$$\int |x|^r d\nu_j = o(M_j^{r/N}). \quad (3.1)$$

Proof. Put $\theta = r/N$. For any $R > 0$, Hölder's inequality on the tail gives

$$\int |x|^r d\nu_j \leq R^r + M_j^\theta \nu_j(|x| > R)^{1-\theta}.$$

Dividing by M_j^θ and taking the upper limit, total-variation convergence gives

$$\limsup_{j \rightarrow \infty} \frac{\int |x|^r d\nu_j}{M_j^{r/N}} \leq \nu(|x| > R)^{1-r/N}.$$

Letting $R \rightarrow \infty$ makes the right side vanish. $\square$

Lemma 3.2. *Let normalized $u_j \rightarrow v$ in $L^2(\mathbb{R}^n)$ and suppose $M_j = \langle u_j, |q|^N u_j \rangle \rightarrow \infty$. Then the position laws of u_j converge to that of v in total variation, and*

$$\mu_\alpha(u_j) = o(M_j^{|\alpha|/N}) \quad (0 < |\alpha| < N). \quad (3.2)$$

Proof. The Cauchy–Schwarz inequality gives

$$\int ||u_j|^2 - |v|^2| \leq \|u_j - v\|_2 \|u_j + v\|_2 \leq 2\|u_j - v\|_2 \rightarrow 0, \quad (3.3)$$

which is total-variation convergence of the position laws. Lemma 3.1 together with $|x^\alpha| \leq |x|^{|\alpha|}$ then gives (3.2). $\square$

4. A COERCIVE NEGATIVE CHANGE-OF-STATE ASYMPTOTIC

Lemma 4.1. *Fix a normalized Schwartz vector v and let normalized Schwartz vectors $u_j \rightarrow v$ in norm satisfy $M_j \rightarrow \infty$. Writing $r_\alpha^{(j)} = r_\alpha(v, u_j)$,*

$$r_\alpha^{(j)} = o(M_j^{|\alpha|/N}) \quad (0 < |\alpha| < N), \quad r_\alpha^{(j)} = -\mu_\alpha(u_j) + o(M_j) \quad (|\alpha| = N). \quad (4.1)$$

Proof. The moments of v are fixed, so for every $0 < |\gamma| < N$ we also have $\mu_\gamma(v) = o(M_j^{|\gamma|/N})$. Induct on $|\alpha|$ in (2.6) using (3.2): each term of the sum is a product of two quantities whose little- o weights add to $|\alpha|$, which gives the first assertion. At total degree N the same recurrence leaves $-\mu_\alpha(u_j)$ as the only contribution of top weight, all other terms being $o(M_j)$. $\square$

Proposition 4.2. *Under the hypotheses of Lemma 4.1,*

$$\langle v, :P(q) :_{u_j} v \rangle \leq -\varepsilon M_j + o(M_j) \rightarrow -\infty. \quad (4.2)$$

Proof. Write $P = P_N + P_{<N}$ with P_N homogeneous of degree N . By linearity of the renormalization, (4.1) gives

$$\langle v, :P(q) :_{u_j} v \rangle = -\langle u_j, P_N(q) u_j \rangle + o(M_j). \quad (4.3)$$

Evaluating the coercivity hypothesis $P(x) \geq \varepsilon |x|^N - k$ at tx , dividing by t^N and letting $t \rightarrow \infty$ gives $P_N(x) \geq \varepsilon |x|^N$, so $\langle u_j, P_N(q) u_j \rangle \geq \varepsilon M_j$. $\square$

5. THE OPPOSING GROUND-STATE BOUND

Lemma 5.1. *Let u be a normalized Schwartz ground state of $H_u = H_0 + \lambda : P(q) :_u$ with $\lambda > 0$. Then for every normalized Schwartz vector v ,*

$$\langle v, : P(q) :_u v \rangle \geq P(0) - \frac{\langle v, H_0 v \rangle}{\lambda}, \quad (5.1)$$

and for the free vacuum u_0 , uniformly in λ ,

$$\langle u_0, : P(q) :_u u_0 \rangle \geq P(0). \quad (5.2)$$

Proof. The expectation normalization in (2.2), including $: 1 :_u = 1$, gives $\langle u, : P(q) :_u u \rangle = P(0)$, so the ground energy satisfies $E_u = \langle u, H_0 u \rangle + \lambda P(0) \geq \lambda P(0)$ because $H_0 \geq 0$. Ground-state minimality against v gives $E_u \leq \langle v, H_0 v \rangle + \lambda \langle v, : P(q) :_u v \rangle$, and combining the two yields (5.1). Taking $v = u_0$ and using $\langle u_0, H_0 u_0 \rangle = 0$ gives (5.2), in which λ no longer appears. $\square$

The two preceding sections now point in opposite directions, and that is the whole mechanism: Proposition 4.2 sends the same expectation to $-\infty$ whenever the top moment escapes, while Lemma 5.1 holds it above a fixed number.

6. AUTOMATIC BOUNDEDNESS OF THE OMITTED MOMENT

Proposition 6.1. *Let $u : [0, L] \rightarrow L^2(\mathbb{R}^n)$ be any normalized nonnegative Schwartz-valued norm-continuous self-consistent branch on a compact coupling interval. Then*

$$\sup_{0 \leq \lambda \leq L} \langle u(\lambda), |q|^N u(\lambda) \rangle < \infty, \quad (6.1)$$

and consequently

$$\sup_{0 \leq \lambda \leq L} |\langle u(\lambda), P(q) u(\lambda) \rangle| < \infty. \quad (6.2)$$

Proof. Suppose the radial top moment were unbounded, so that there are $\lambda_j \in [0, L]$ with $M_j = \langle u(\lambda_j), |q|^N u(\lambda_j) \rangle \rightarrow \infty$. Passing to a subsequence, $\lambda_j \rightarrow \lambda_*$ by compactness, and norm continuity gives $u_j := u(\lambda_j) \rightarrow v := u(\lambda_*)$ in L^2 . The values $\lambda_j = 0$ cannot witness escape, since $u(0)$ is a fixed vector, so we may take $\lambda_j > 0$. By Proposition 4.2,

$$W_j := \langle v, : P(q) :_{u_j} v \rangle \rightarrow -\infty. \quad (6.3)$$

If $\lambda_* > 0$ then eventually $\lambda_j \geq \lambda_*/2$, and (5.1) with the fixed test vector v gives $W_j \geq P(0) - 2\langle v, H_0 v \rangle / \lambda_*$, contradicting (6.3). If $\lambda_* = 0$ then continuity and uniqueness of the normalized nonnegative free ground state give $v = u(0) = u_0$, and (5.2) gives $W_j \geq P(0)$, again contradicting (6.3). This proves (6.1), and (6.2) follows from $|P(x)| \leq C_P(1 + |x|^N)$. $\square$

7. UNIFORM UNIQUENESS ON EVERY SUFFICIENTLY SHORT INTERVAL

We use the finite-dimensional cone inverse construction of Baez and Zhou [12, Theorem 1 and Lemmas 8–9], keeping track of the interval quantifier.

Let $\mathcal{P}_N$ be the finite-dimensional space of real polynomials of degree at most N , and let $\mathcal{C}_+ \subset \mathcal{P}_N$ consist of those whose degree- N homogeneous form is strictly positive on the unit sphere; this is an open convex cone contained in the cone of polynomials bounded below. For $Q \in \mathcal{C}_+$ let $v(Q)$ be the normalized nonnegative ground state of $H_0 + Q(q)$ and define $T(Q)$ by

$$Q(q) = : T(Q)(q) :_{v(Q)}. \quad (7.1)$$

The cited source proves that T is analytic on $\mathcal{C}_+$ and that $T(Q) \rightarrow 0$ and $dT(Q) \rightarrow I$ as $Q \rightarrow 0$ within the cone, and its cone inverse lemma supplies $\Lambda > 0$ and a continuous curve

$$y : [0, \Lambda] \rightarrow \mathcal{C}_+ \cup \{0\}, \quad T(y(\lambda)) = \lambda P, \quad y(0) = 0. \quad (7.2)$$

That lemma's uniqueness argument applies verbatim on every restricted interval $[0, L]$ with $0 < L \leq \Lambda$: near zero the contraction estimate forces any continuous solution of (7.2) to equal y , and if equality first failed later, local invertibility of T at the common first-disagreement point would give a contradiction. So $y|_{[0, L]}$ is the unique continuous cone-valued solution on each such interval.

Lemma 7.1. *Fix $0 < L \leq \Lambda$ and let $w : [0, L] \rightarrow L^2(\mathbb{R}^n)$ be any branch allowed in Theorem 1.1. Set*

$$Q_w(0) = 0, \quad Q_w(\lambda) = \lambda : P(q) :_{w(\lambda)} \quad (\lambda > 0). \quad (7.3)$$

Then Q_w takes values in $\mathcal{C}_+ \cup \{0\}$, is continuous coefficientwise on $[0, L]$ including at $\lambda = 0$, and satisfies $T(Q_w(\lambda)) = \lambda P$.

Proof. The leading form of $Q_w(\lambda)$ is λP_N , so $Q_w(\lambda) \in \mathcal{C}_+$ for $\lambda > 0$, and self-consistency with (7.1) gives $w(\lambda) = v(Q_w(\lambda))$ and $T(Q_w(\lambda)) = \lambda P$.

For continuity, write $B = \sup_{0 \leq \lambda \leq L} \langle w(\lambda), |q|^N w(\lambda) \rangle$, which is finite by Proposition 6.1. Norm continuity gives total-variation continuity of the position laws by (3.3), and for $0 \leq r < N$ the uniform tail estimate

$$\int_{|x| > R} |x|^r |w(\lambda, x)|^2 dx \leq R^{r-N} B \quad (7.4)$$

holds. Truncating at R and letting $R \rightarrow \infty$ shows every moment of total degree below N is continuous in λ . By (2.6), every nonconstant coefficient of $: P :_{w(\lambda)}$ depends only on moments of total degree at most $N - 1$, so the nonconstant part Q_w^{nc} is continuous, including at zero.

For the constant coefficient c_w , adding a scalar to a potential changes neither its ground state nor any positive-degree renormalized product, so $T(R + c) = T(R) + c$. With $T(Q_w(\lambda)) = \lambda P$ this gives

$$c_w(\lambda) = \lambda P(0) - [T(Q_w^{\text{nc}}(\lambda))]_0, \quad (7.5)$$

where $[\cdot]_0$ is the constant coefficient. Continuity of T on the cone, together with $T(Q) \rightarrow 0$ as $Q \rightarrow 0$, makes c_w continuous on $[0, L]$. $\square$

Proof of Theorem 1.1. Let Λ be as in (7.2) and fix $0 < \lambda_0 \leq \Lambda$. Existence of a branch on $[0, \lambda_0]$ follows by restricting y and setting $u(\lambda) = v(y(\lambda))$.

For uniqueness, let w be any branch allowed by the statement. By Lemma 7.1, Q_w is a continuous cone-valued solution of (7.2) on $[0, \lambda_0]$, so restricted-interval cone uniqueness gives $Q_w = y|_{[0, \lambda_0]}$ and hence $w(\lambda) = v(y(\lambda))$ for every λ .

No bound on $\langle w(\lambda), P(q)w(\lambda) \rangle$ was assumed at any point: the one used in Lemma 7.1 was supplied by Proposition 6.1 from norm continuity and self-consistency alone. $\square$

8. CONCLUSION

Norm continuity, coercivity, and ground-state minimality together prevent escape of the interaction moment. The conclusion is local in coupling and finite dimensional; it neither supplies maximal continuation nor removes continuity at the free-vacuum endpoint. Within that boundary, the formerly assumed moment condition is redundant.

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